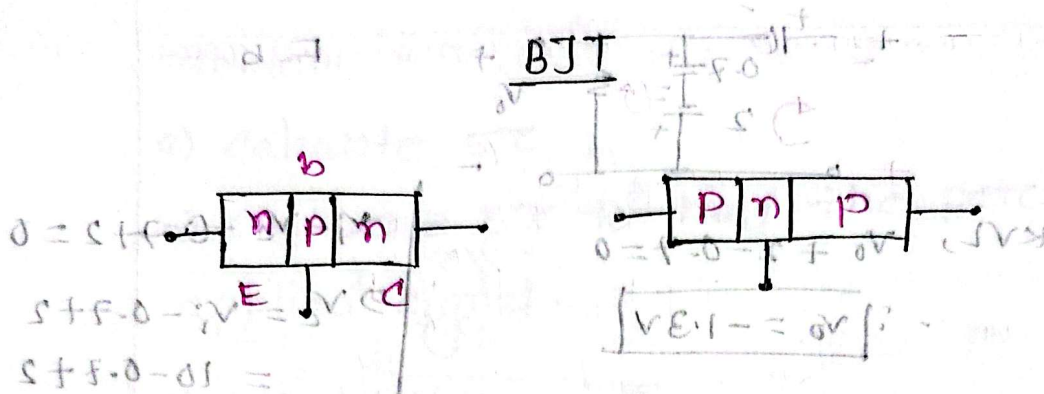




width: $C > E > B$

doping: $E > C > B$

Transistor $\Rightarrow$ Transistor + Resistor

$J_1 \rightarrow F.b$

$J_2 \rightarrow R.b$

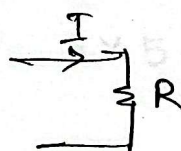
} Amplifier

$J_1 \rightarrow F.b \rightarrow Res = 0$

$J_2 \rightarrow R.b \rightarrow Res = \infty$

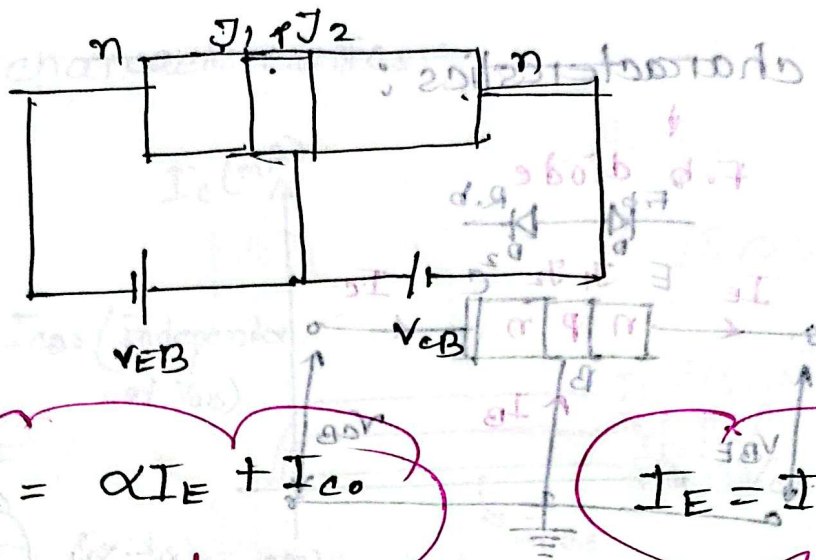


$$V_i = I \times r$$



$$V_o = I \times R$$

$\therefore V_i < V_o \rightarrow$ amplification



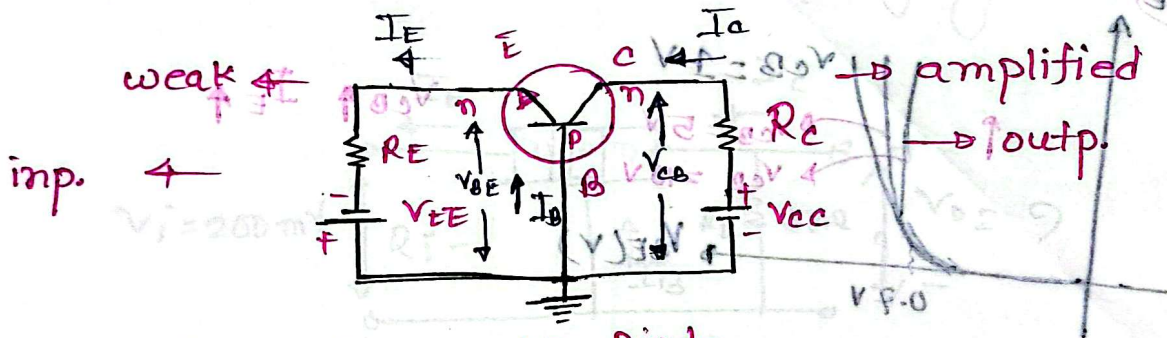
$$I_C = \alpha I_E + I_{C0}$$

$$I_E = I_B + I_C$$

Common Base Configuration?

$$V_{BE} = V_{EE} \text{ if } R_E \times$$

$$V_{CB} = V_{CC} \text{ if } R_C \times$$



$J_1 \rightarrow F.B \rightarrow EB \text{ Diode}$

$J_2 \rightarrow R.B \rightarrow BC$

inp. C $\rightarrow I_E$ V_s V_{BE}

out. C $\rightarrow I_C$ V_s V_{CB}

$$\alpha = \frac{I_C}{I_E}$$

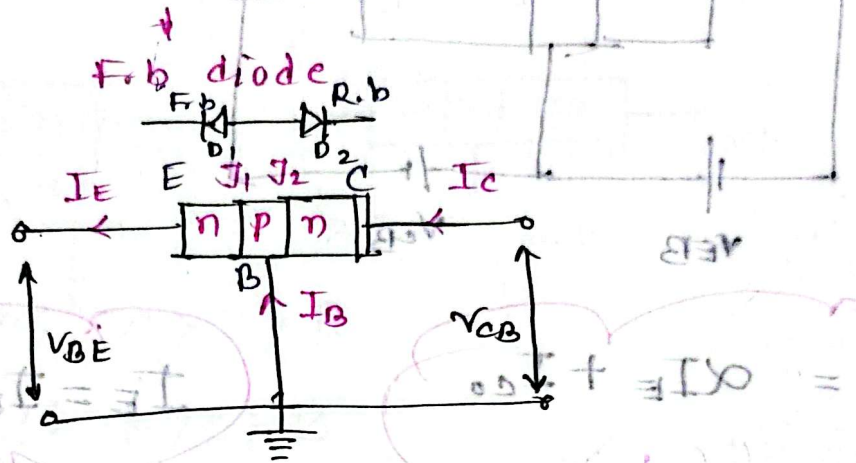
$$I_E = I_B + I_C$$

$$I_C = \alpha I_E + I_{C0}$$

$$I_C = \alpha I_E + I_{B0}$$

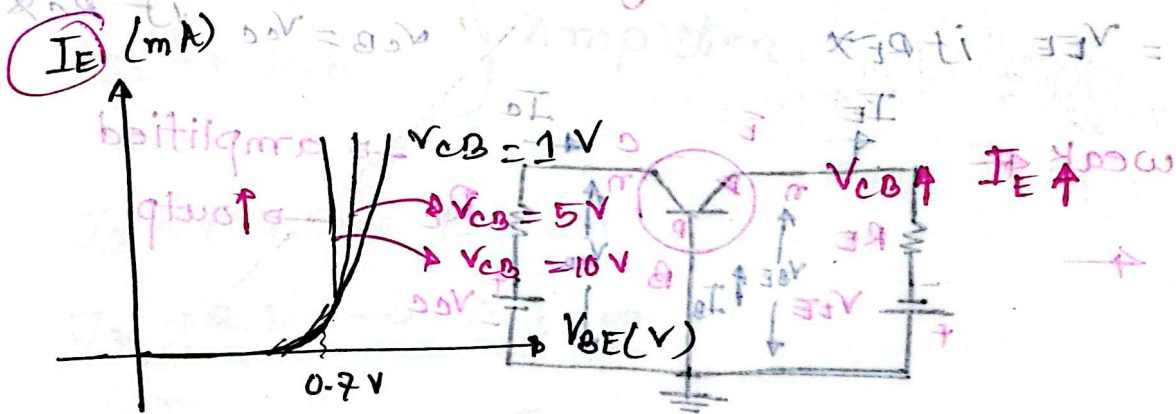
$\alpha = \text{amplification factor}$ $I_B = (1 - \alpha) I_E$

Input characteristics :

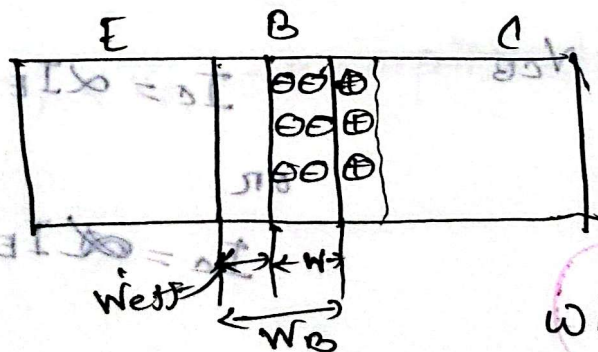


inp. $I = I_E$, inp. $V = V_{BE}$

Outp. $I = I_C$, Outp. $V = V_{CB}$



Early effect : base width modulation



$$W_B = \text{width of base}$$

$$W_B = W_{eff} + W$$

$$\Rightarrow W_{eff} = W_B - W$$

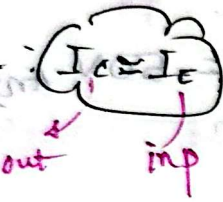
when $V_{CB} \uparrow \Rightarrow W \uparrow \rightarrow W_{eff} \downarrow$

output characteristics:

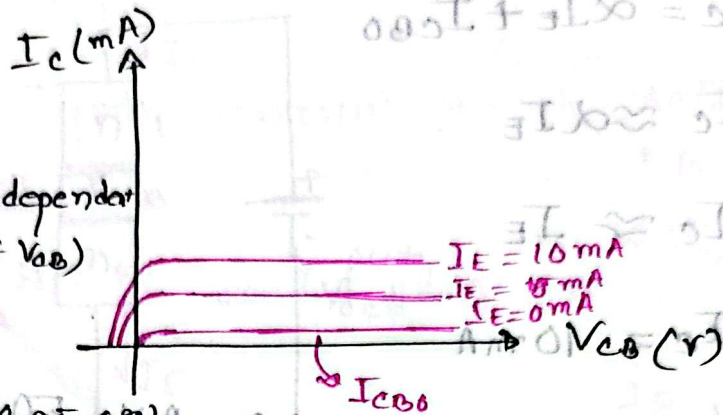
(p.b)

$$I_C = \alpha I_E + I_{CBO} \text{ (independent of } V_{CE})$$

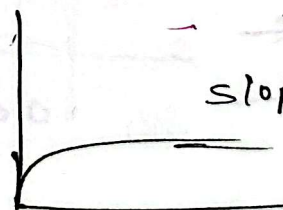
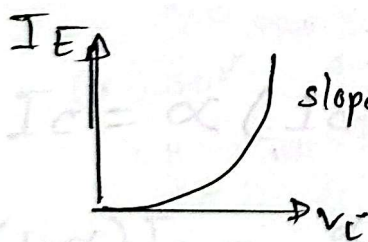
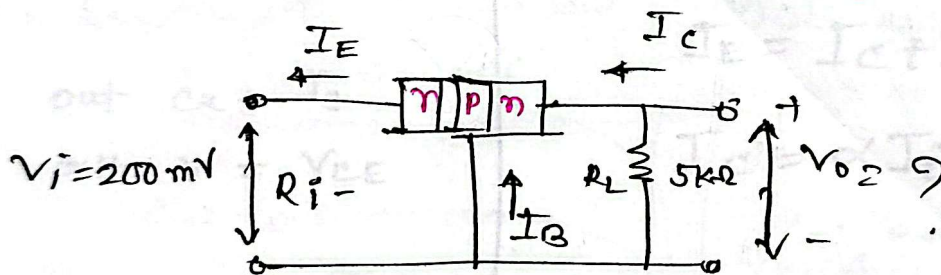
$$I_C \approx \alpha I_E$$



$$(\alpha = 0.95 - 0.99)$$



Transistor amplifying action



$$R_o = 1 \text{ M}\Omega = 100 \text{ k}\Omega$$

$$R_i = 20 \Omega$$

$$V_i = I_E R_i$$

$$\Rightarrow I_E = \frac{V_i}{R_i} = \frac{200 \text{ mV}}{20} = 10 \text{ mA}$$

Then

$$I_C = \alpha I_E + I_{CBO}$$

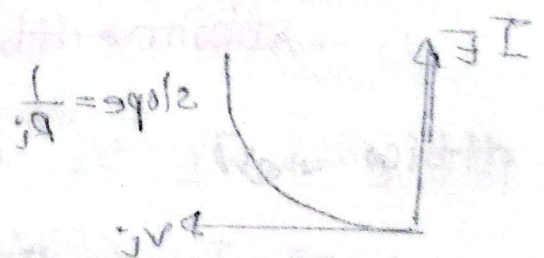
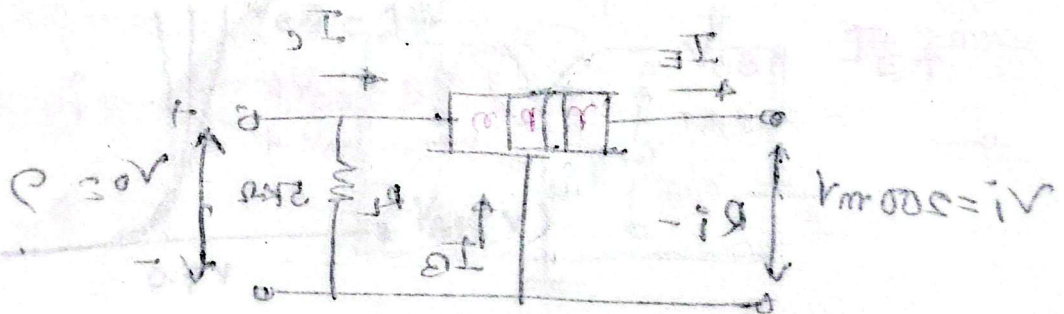
$$\Rightarrow I_C \approx \alpha I_E$$

$$\Rightarrow I_C \approx I_E$$

$$\therefore I_C = 10 \text{ mA}$$

$$\therefore V_o = I_C \times R_L = 10 \text{ mA} \times 50 \text{ k}\Omega = 500 \text{ V}$$

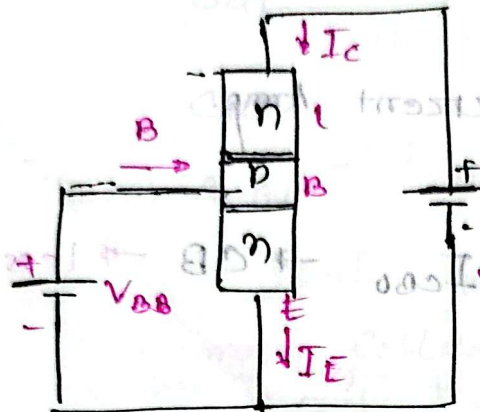
$$\text{Amp} = \frac{V_o}{V_i} = \frac{500}{10 \text{ mV}} = 50$$



$$I_E = \frac{V_i}{R_i} = \frac{10 \text{ mV}}{200 \Omega} = 50 \text{ }\mu\text{A}$$

Common Emitter

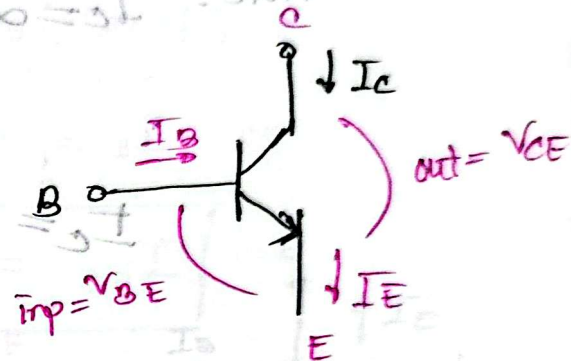
(Imp)



Active

EB → f.

C-B → R.



$$\text{inp. cu} = I_B$$

$$\text{inp. } v = V_{BE}$$

$$\text{out cu} = I_C$$

$$\text{out } v = V_{CE}$$

$$I_E = I_C + I_B \quad \leftarrow (i)$$

$$I_C = \alpha I_E + I_{CBO} \quad \leftarrow (ii)$$

$$\rightarrow \alpha < 1$$

$$\rightarrow 50 \leq \beta \leq 400$$

$$I_C = \alpha (I_C + I_B) + I_{CBO}$$

$$\Rightarrow (1-\alpha)I_C = \alpha I_B + I_{CBO}$$

$$\Rightarrow I_C = \frac{\alpha}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CBO}$$

$$\Rightarrow I_C = \beta I_B + (\beta+1) I_{CBO}$$

$$\Rightarrow I_C = \beta I_B + (\beta+1) I_{CBO}$$

$$\Rightarrow I_C = \beta I_B ; I_{CBO} \ll \beta I_B$$

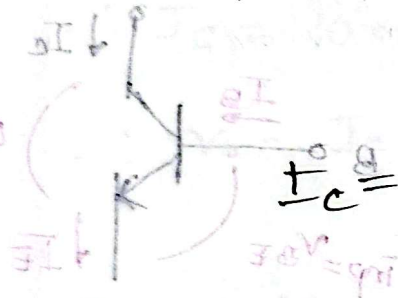
$$(\beta+1) I_{CBO} = I_{CEO}$$

$$\therefore \beta = \frac{I_C}{I_B}$$

β = Current amp fact

CE Conf. $\rightarrow$ Current Amp.

Note: $I_C = \alpha I_E + I_{CBO}$ $\rightarrow$ CB $\rightarrow$ less conf.



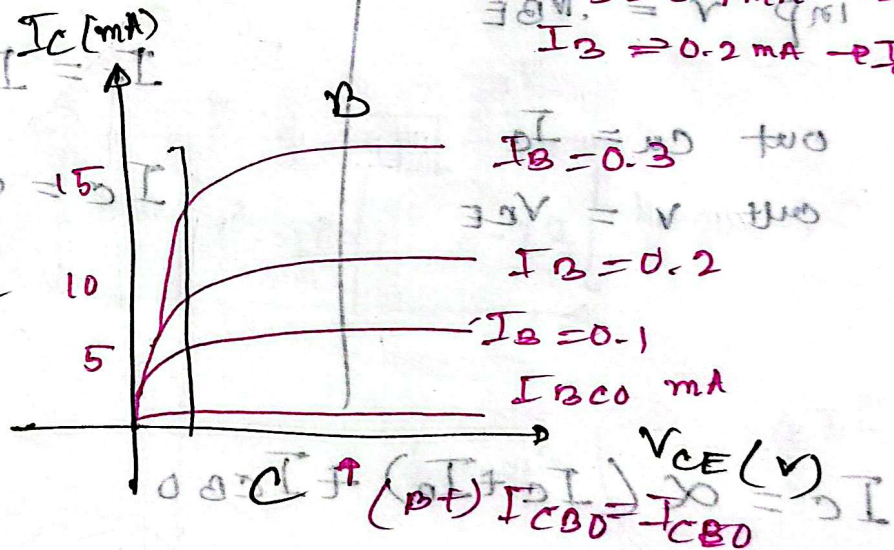
$$I_C = \beta I_B + (\beta + 1) I_{CBO} \rightarrow \text{CE} \rightarrow \text{high. Conf.}$$

Output characteristics:

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

$$I_B = 0$$

$$I_C = (\beta + 1) I_{CBO}$$



$$\beta = \frac{10 - 5}{0.2 - 0.1} = 50$$

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

Case 1

A

B

C

Case 2

Sat

Cut-

Activ

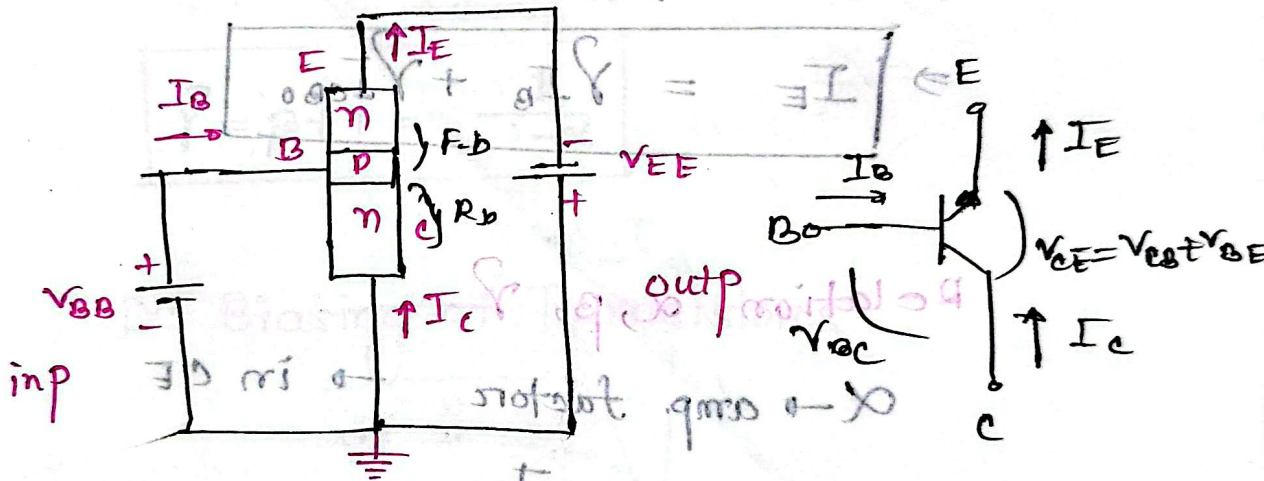
$$(i) - \beta I_B + I_C = I_E$$

$$(ii) - \beta I_B + I_C = I_E$$

$$\beta I_B + I_C = I_E$$

$$\beta I_B + I_C = I_E (\beta + 1)$$

Common Collector



out $\therefore I_E, V_S, V_{CE}$ for various level I_B

$$I_C = \alpha I_E$$

$$\Rightarrow I_C \approx I_E$$

* out. chara. of CE = output. char. of CC

$$I_E = I_C + I_B \quad \text{--- (i)}$$

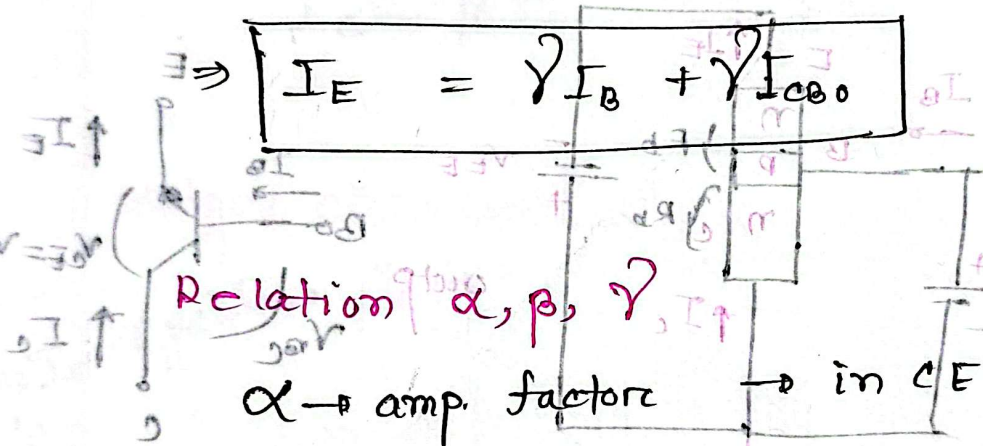
$$I_C = \alpha I_E + I_{CBO} \quad \text{--- (ii)}$$

$$I_E = \alpha I_E + I_{CBO} + I_B$$

$$\Rightarrow (1 - \alpha) I_E = I_B + I_{CBO}$$

$$\Rightarrow I_E = \frac{1}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO}$$

$$\Rightarrow I_E = \gamma I_B + \gamma I_{CBO}$$



$\alpha \rightarrow$ amp. factor

$$\alpha \rightarrow \alpha_{DC} = \frac{I_C}{I_E}$$

$$\alpha \rightarrow \alpha_{AC} = \frac{\Delta I_C}{\Delta I_E} \quad \left| \begin{array}{l} V_{CE} = \text{const.} \\ V_{CB} = \text{const.} \end{array} \right.$$

$$\beta \rightarrow D_C = \frac{I_C}{I_B}$$

$$\beta \rightarrow A_C = \frac{\Delta I_C}{\Delta I_B} \quad \left| \begin{array}{l} V_{CE} = \text{const.} \\ V_{BE} = \text{const.} \end{array} \right.$$

cc to. output. input = 30 to. output. two *

$$\gamma \rightarrow D_C = \frac{I_E}{I_B}$$

$$\gamma \rightarrow A_C = \frac{\Delta I_E}{\Delta I_B} \quad \left| \begin{array}{l} V_{CE} = \text{const.} \\ V_{BE} = \text{const.} \end{array} \right.$$

$$I_E = \frac{I_C}{\alpha} + I_B$$

$$\Rightarrow \frac{I_E}{I_B} = \frac{I_C}{I_B} + 1$$

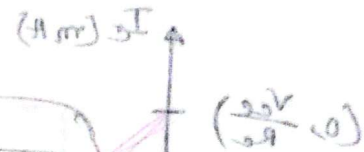
$$\Rightarrow \boxed{\gamma = \beta + 1}$$

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}$$

$$\gamma = \frac{\alpha}{1-\alpha} + 1$$

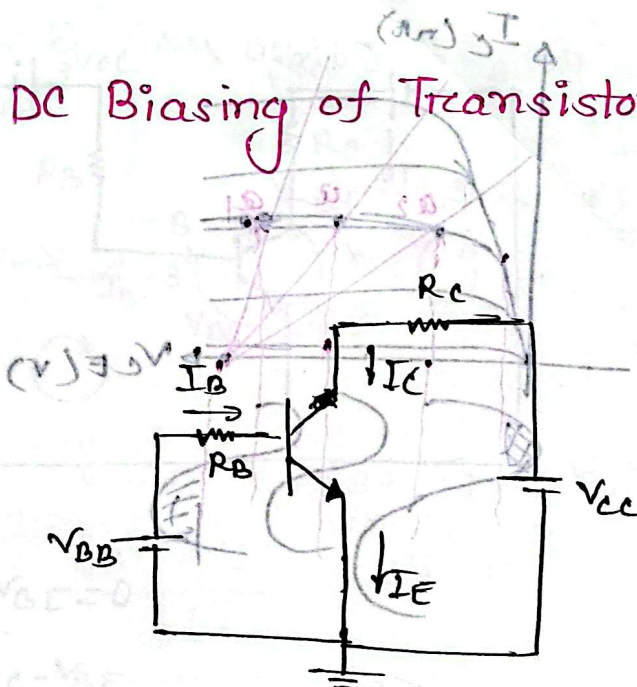
$$\boxed{\gamma = \frac{1}{1-\alpha}}$$

$$\boxed{\gamma = \beta + 1 = \frac{1}{1-\alpha}}$$



$$I_C = \frac{\beta}{\beta+1} I_E$$

DC Biasing of Transistors

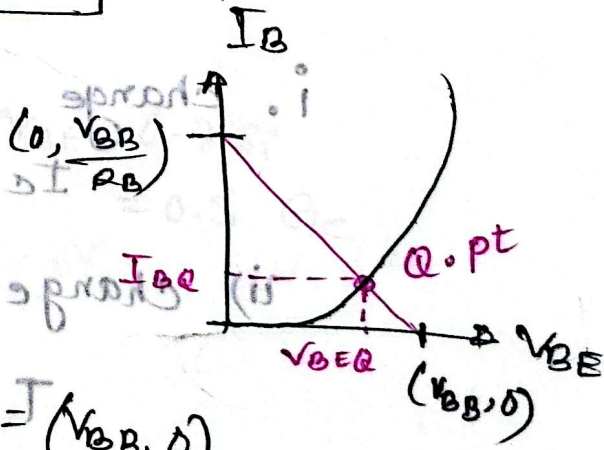


operating point

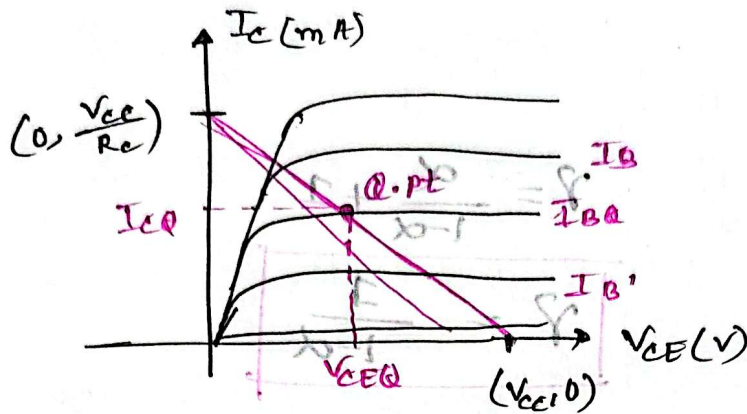
$$V_{BB} - I_B R_B - V_{BE} = 0$$

$$P_1 = (0, \frac{V_{BB}}{R_B})$$

$$P_2 = (V_{BE}, 0)$$



$$V_{CC} - I_C R_C - V_{CE} = 0$$



$$V_{CE} = 0$$

$$I_C = 0$$

$$I_C = \frac{V_{CC}}{R_C}$$

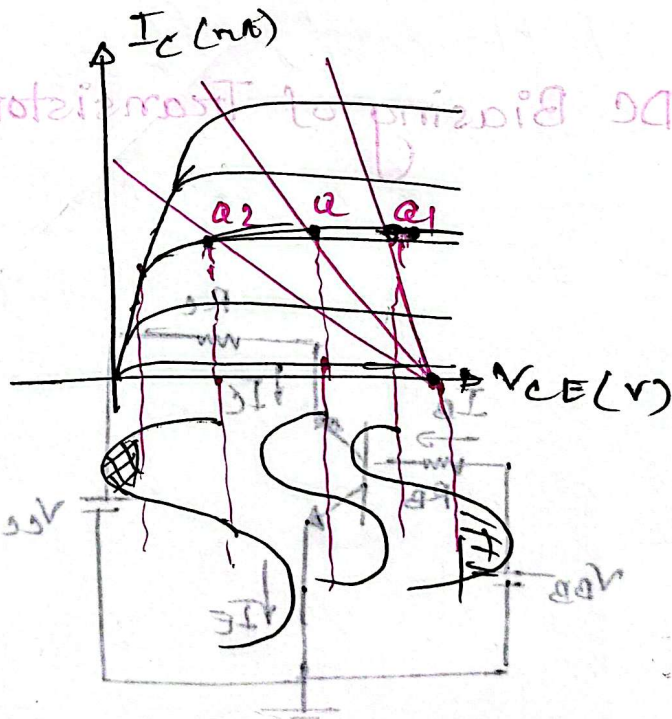
$$\frac{V_{CE}}{V_{CE0}} = \frac{I_C}{I_{C0}} = 1 + \beta = \gamma$$

$$\frac{\partial I_C}{\partial I_B} = \beta$$

$$\frac{\partial I_C}{\partial I_B} = \frac{\partial I_C}{\partial I_B} = \beta$$

$$\text{slope} = -\frac{1}{R_C} \gamma$$

$$\frac{\gamma}{\gamma - 1} = \beta$$



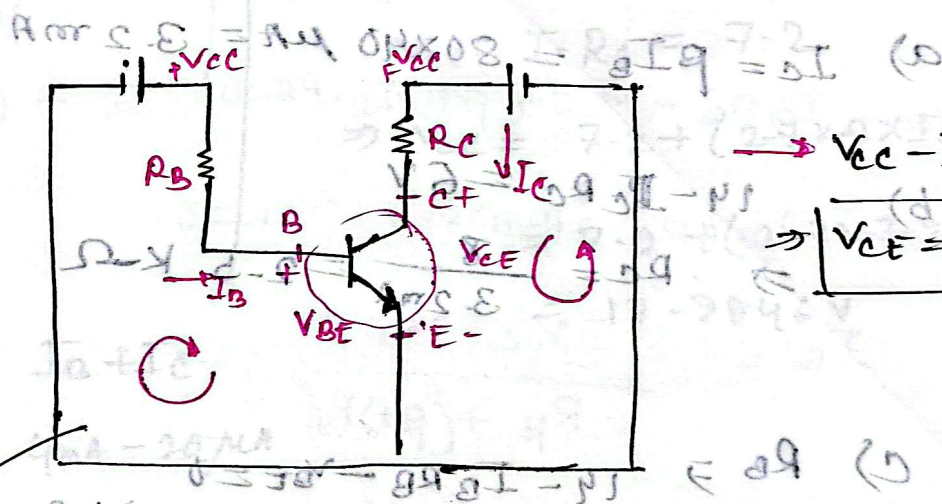
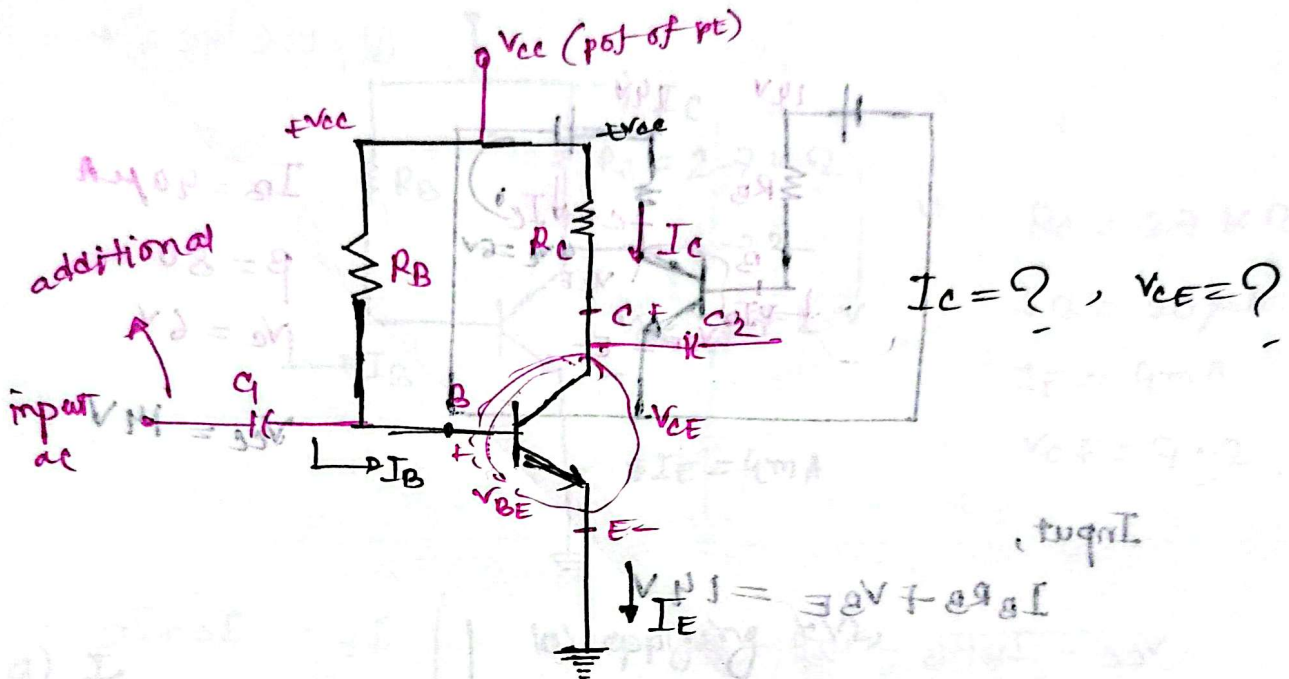
i. change in β :

$$I_C = \beta I_B, \beta \uparrow \rightarrow I_C \uparrow$$

ii) change in I_B :

$$I_C \uparrow \rightarrow I_C \uparrow \quad \left(\frac{\partial I_C}{\partial I_B} = \beta \right)$$

Fixed Bias Configuration of a transistor (Base Bias)



$$V_{CC} - I_B R_B - V_{BE} = 0$$

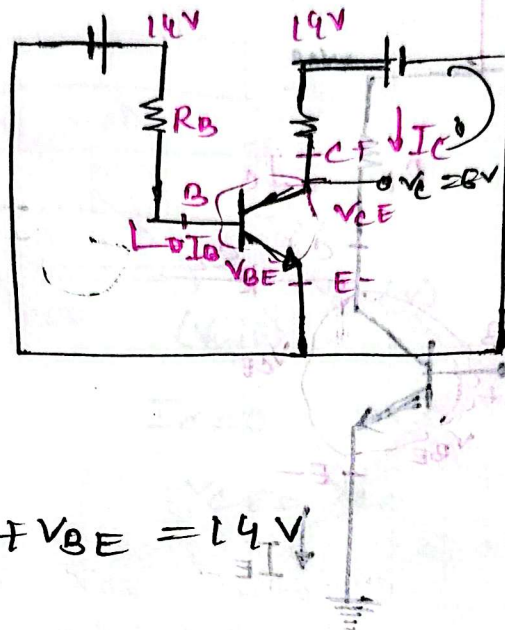
$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\therefore I_C = \beta I_B = \frac{\beta (V_{CC} - V_{BE})}{R_B}$$

$$V_{BE} = 0.7 \text{ V}$$

$$= 0.3 \text{ V}$$

Example:



$$I_B = 40 \mu A$$

$$\beta = 80$$

$$V_C = 6V$$

$$V_{CE} = 14V$$

Input,

$$I_B R_B + V_{BE} = 14V$$

$$a) I_C = \beta I_B = 80 \times 40 \mu A = 3.2 \text{ mA}$$

$$b) 14 - I_C R_C = 6V$$

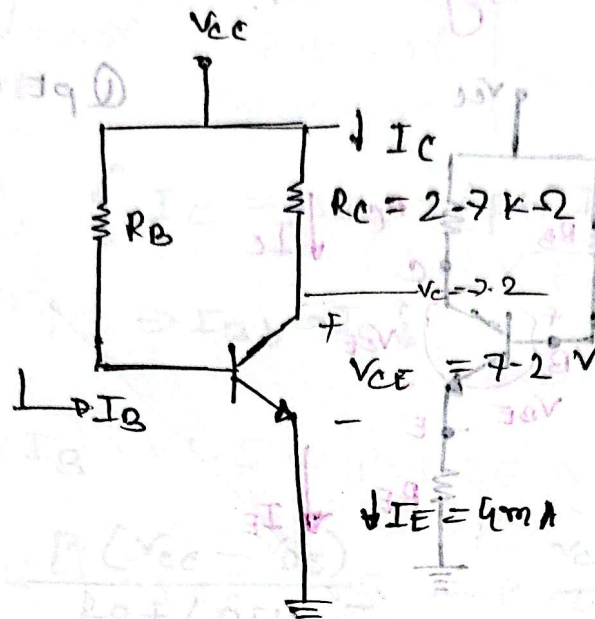
$$\Rightarrow R_C = \frac{8}{3.2 \text{ mA}} = 2.5 \text{ K}\Omega$$

$$c) R_B \Rightarrow 14 - I_B R_B - V_{BE} = 0$$

$$\Rightarrow R_B = \frac{14 - V_{BE}}{I_B} = \frac{13.3}{40 \mu A} = 332.5 \text{ K}\Omega$$

$$d) V_{CE} = V_C - V_E = 6 - 0 = 6V$$

problem:



$$R_C = 2.7 \text{ k}\Omega$$

$$I_B = 20 \mu\text{A}$$

$$I_E = 4 \text{ mA}$$

$$V_{CE} = 7.2 \text{ V}$$

a) I_C

b) V_{CE}

c) β

d) R_B

applying KVL

$$V_{CC} - I_C R_C = 7.2$$

$$\Rightarrow V_{CC} = 7.2 + (2.7 \text{ k}\Omega \times I_C)$$

$$17.9046 = 7.2 + (2.7 \times 3.998)$$

$$a) I_E = I_B + I_C$$

$$\Rightarrow I_C = 4 \text{ mA} - 20 \mu\text{A}$$

$$\therefore I_C = 3.98 \text{ mA}$$

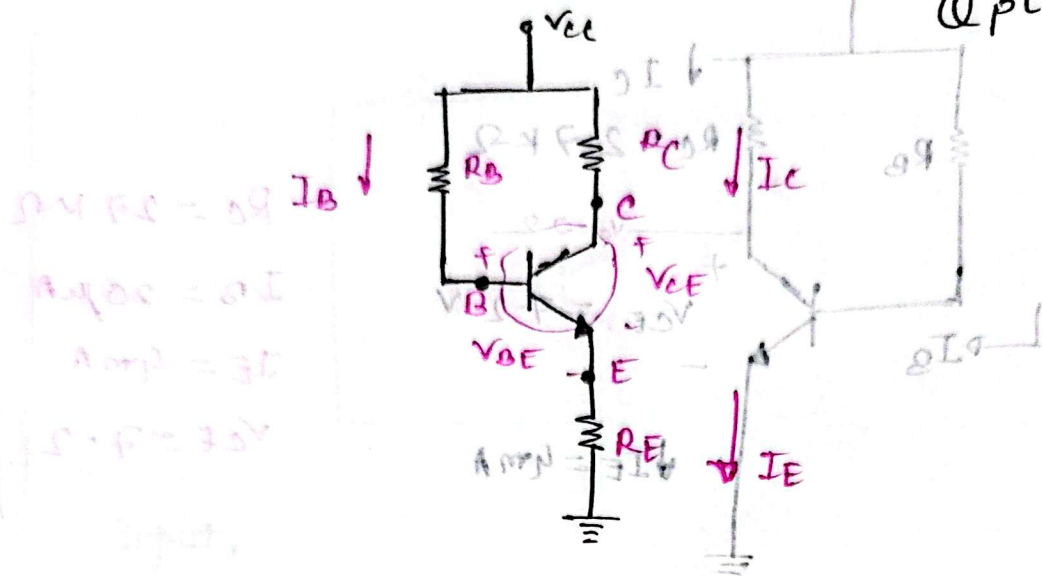
$$c) \beta = \frac{I_C}{I_B} = \frac{3.998}{20 \times 10^{-3}} = 199.9$$

$$d) V_{CC} - I_B R_B - V_{BE} = 0$$

$$\Rightarrow 17.9046 - (20 \mu\text{A}) R_B - 0.7 = 0$$

$$\Rightarrow R_B = (862.03 \text{ k}\Omega)$$

Emitter Bias Configuration:



$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = I_C + I_B = \beta I_B + I_B = (\beta + 1) I_B$$

$$\Rightarrow V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$\Rightarrow V_{CC} - I_B (R_B + (\beta + 1) R_E) - V_{BE} = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$

$$\Rightarrow I_C = \beta \left(\frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \right)$$

for outp,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - I_C (R_C + R_E) - V_{CE} = 0$$

$$\Rightarrow V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Ad by using R_E

i) $I_C \Rightarrow Q.Pt$

$V_{CC} = 12V$
 $V_{BE} = 0.7V$
 $V_{CE} = 3V$

$I_{BQ} = 50 \mu A$
 $\beta = 80$

$\Rightarrow \uparrow \uparrow I_C \uparrow \left\{ I_C = \beta I_B + (1 + \beta) I_{CBO} \right\}$

$\Rightarrow -I_E \uparrow \Rightarrow I_B \downarrow \Rightarrow I_C \uparrow$

$V_{CE} = 3V - 0.7V$

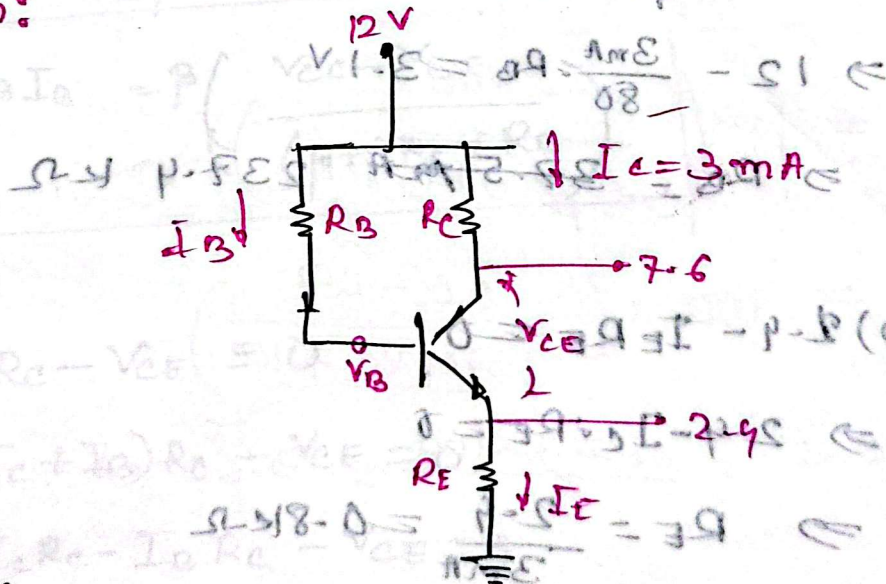
ii) $I_C = (\beta I_B)$ $V_{BE} = 0.7V + 0.7V = 1.4V$

$I_C = \frac{\beta (V_{CC} - V_{BE})}{R_B + (\beta + 1) R_E} \approx \frac{V_{CC} - V_{BE}}{R_E} \text{ ind. of } \beta$

$\Rightarrow R_E \uparrow \uparrow, R_B \downarrow \downarrow$

$0.7V = 0.7V - 0.7V$

problem:



a) R_C

b) R_E

c) R_B

d) V_{CE}

e) V_B

$$\begin{array}{l|l} I_C = 3 \text{ mA} & V_C = 7.6 \\ \beta = 80 & V_E = 2.4 \end{array}$$

$$V_{CC} = 12 \text{ V}$$

$$c) V_{BE} = 0.7 \text{ V (Si)}$$

$$V_B - V_E = 0.7$$

$$\Rightarrow V_B = 0.7 + 2.4 = 3.1 \text{ V (Ans)}$$

$$d) V_{CE} = V_C - V_E = 7.6 - 2.4 = 5.2 \text{ V (Ans)}$$

$$c) 12 - I_B R_B = V_B$$

$$\Rightarrow 12 - \frac{I_C}{\beta} \cdot R_B = 3.1 \text{ V}$$

$$\Rightarrow 12 - \frac{3 \text{ mA}}{80} \cdot R_B = 3.1 \text{ V}$$

$$\Rightarrow R_B = 37.5 \text{ k}\Omega$$

$$b) 2.4 - I_E R_E = 0$$

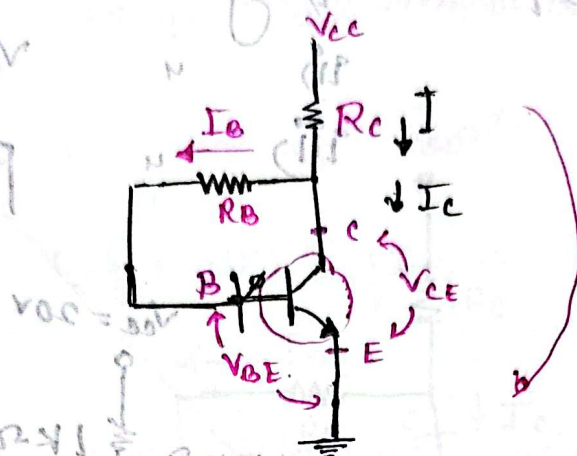
$$\Rightarrow 2.4 - I_C \cdot R_E = 0$$

$$\Rightarrow R_E = \frac{2.4}{3 \text{ mA}} = 0.8 \text{ k}\Omega$$

$$a) 12 - I_C R_C = 7.6$$

$$\Rightarrow R_C = \frac{12 - 7.6}{I_C} = 1.47 \text{ k}\Omega$$

Collector Feedback biasing



$$I = I_C + I_B$$

Applying KVL (inp)

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0$$

$$\Rightarrow V_{CC} - (\beta I_B + I_B)R_C - I_B R_B - V_{BE} = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{(\beta + 1)R_C + R_B}$$

$$I_C = \beta I_B = \beta \left(\frac{V_{CC} - V_{BE}}{(\beta + 1)R_C + R_B} \right)$$

(outp)

$$V_{CC} - I_C R_C - V_{CE} = 0$$

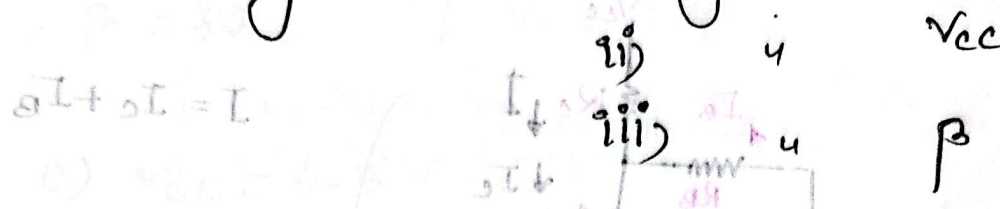
$$\Rightarrow V_{CC} - (I_C + I_B)R_C - V_{CE} = 0$$

$$\Rightarrow V_{CC} - I_C R_C - I_B R_C - V_{CE} = 0$$

$$V_{CE} = \frac{V_{CC} - (I_C + I_B)R_C}{1}$$

$$A_{mid} = 0.0 = 0$$

adv. stabilizing Q point $\rightarrow$ i) against var T



problem:

$$\beta = 100$$

$$V_{CC} = 20 \text{ V}$$

$$R_B = 100 \text{ k}\Omega$$

$$R_C = 1 \text{ k}\Omega$$

$$I_C = ? , V_{CE} = ?$$

$$I_C = \beta \left(\frac{V_{CC} - V_{BE}}{(\beta + 1)R_B + R_B} \right)$$

$$= 100 \left(\frac{20 - 0.7}{100 \times 1 + 100} \right)$$

$$\therefore I_C = \frac{19.3 \times 100}{201} = 9.60 \text{ mA}$$

$$V_{CE} = V_{CC} - (I_B + I_C)R_C$$

$$= 20 - (9.60 + 0.096)1$$

$$\therefore V_{CE} = 10.304 \text{ V (Ans.)}$$

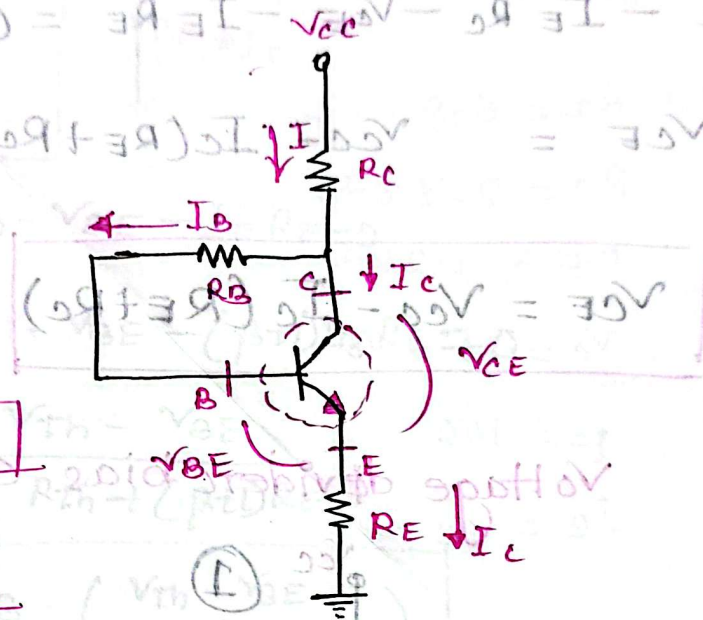
$$\beta = \frac{I_C}{I_B}$$

$$\Rightarrow I_B = \frac{I_C}{\beta}$$

$$= \frac{9.60}{100}$$

$$\therefore I_B = 0.096 \text{ mA}$$

Collector feedback biasing with Emitter Resistance



And

$$I_E = I_C + I_B = (\beta + 1)I_B$$

$$I = I_C + I_B$$

KVL in inp

$$V_{CC} - I_B R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - (I_B + I_C) R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - (\beta + 1)I_B R_C - I_B R_B - V_{BE} - (\beta + 1)I_B R_E = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)(R_C + R_E)}$$

$$I_C = \frac{\beta (V_{CC} - V_{BE})}{R_B + (\beta + 1)(R_C + R_E)}$$

where $R_B \ll (\beta + 1)(R_C + R_E)$

outp:

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\Rightarrow V_{CE} = V_{CC} - (I_B + I_C) R_C - I_E R_E$$

$$\Rightarrow V_{CE} = V_{CC} - (I_B + I_C) R_C - (I_B + I_C) R_E$$

$$\Rightarrow V_{CE} = V_{CC} - (I_B + I_C)(R_C + R_E)$$

output:

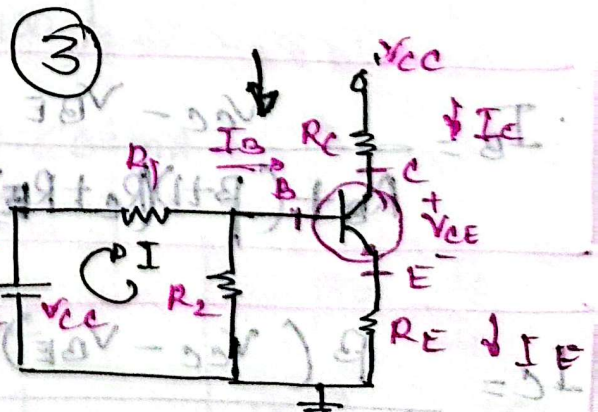
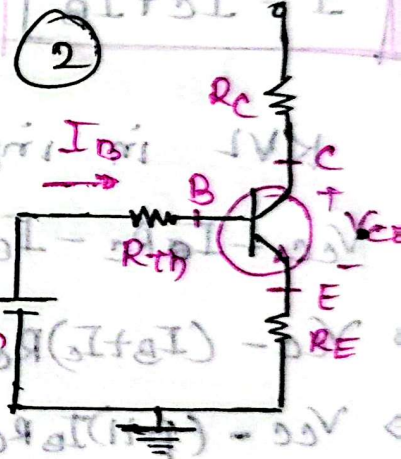
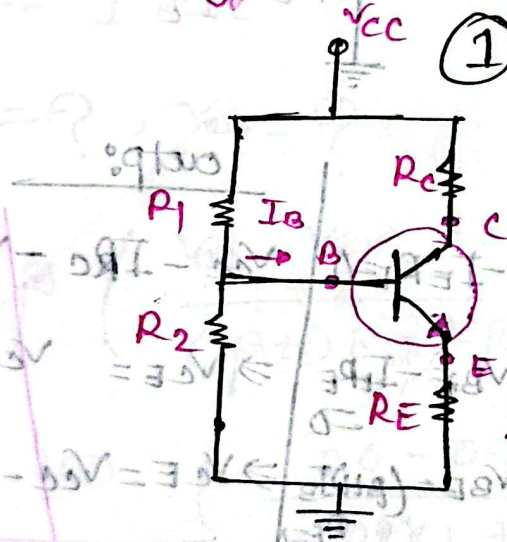
$$V_{CC} - (I_B + I_C) R_C - V_{CE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - I_E R_C - V_{CE} - I_E R_E = 0$$

$$\Rightarrow V_{CE} = V_{CC} - I_C (R_E + R_C) ; I_E \approx I_C$$

$$\therefore \boxed{V_{CE} = V_{CC} - I_C (R_E + R_C)}$$

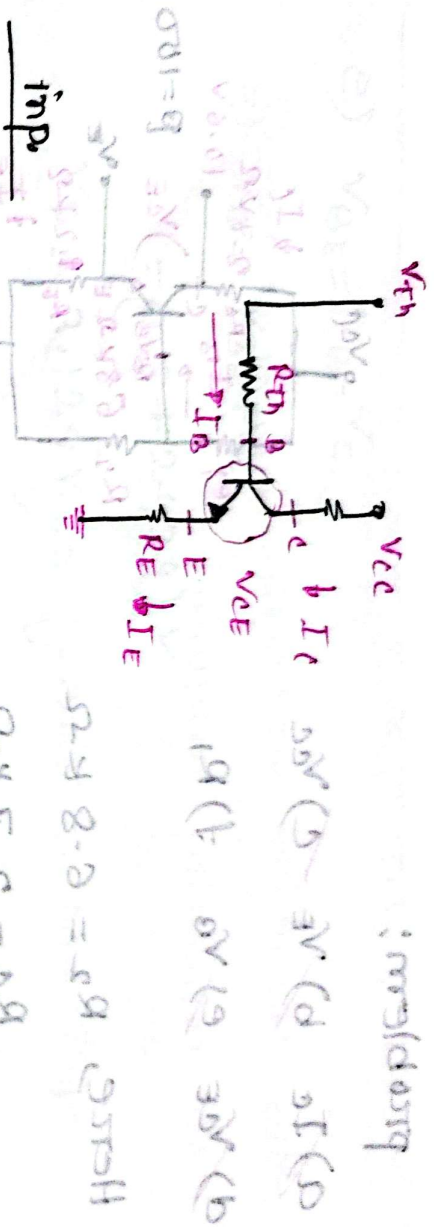
Voltage divider bias configuration



$$\therefore \boxed{R_{Th} = \frac{R_1 R_2}{R_1 + R_2}}$$

$$I = \frac{V_{CC}}{R_1 + R_2}$$

$$V_{Th} = I R_2 = \boxed{\frac{R_2 \cdot V_{CC}}{R_1 + R_2} = V_{Th}}$$



inp: $V_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$
 $9V - I_B (10k + 20k) - 0.7V - I_E (1k) = 0$
 $9V - I_B (30k) - 0.7V - I_E (1k) = 0$
 $8.3V = I_B (30k) + I_E (1k)$
 $8.3V = I_B (30k) + I_B (\beta + 1) (1k)$
 $8.3V = I_B (30k + 1k(\beta + 1))$
 $8.3V = I_B (31k + \beta k)$
 $I_B = \frac{8.3V}{31k + \beta k}$

$\Rightarrow V_{th} - I_B R_{th} - V_{BE} - (\beta + 1) I_B R_E = 0$
 $9V - I_B (30k) - 0.7V - (\beta + 1) I_B (1k) = 0$
 $8.3V = I_B (31k + \beta k)$
 $I_B = \frac{8.3V}{31k + \beta k}$

$\Rightarrow I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}$
 $I_B = \frac{9V - 0.7V}{30k + (\beta + 1) 1k}$
 $I_B = \frac{8.3V}{31k + \beta k}$

$\therefore I_C = \beta \left(\frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E} \right)$
 $I_C = \frac{\beta (V_{th} - V_{BE})}{R_{th} + (\beta + 1) R_E}$
 $I_C = \frac{\beta (8.3V)}{31k + \beta k}$
 $I_C = \frac{8.3V \beta}{31k + \beta k}$

if $R_{th} \ll (\beta + 1) R_E$; I_C independent of β
 $I_C \approx \frac{V_{th} - V_{BE}}{R_E} = \frac{8.3V}{1k} = 8.3mA$

outp: $V_{CE} = V_{CC} - I_C R_C - I_E R_E = 9V - 8.3mA (2k + 1k) = 9V - 16.6V = -7.6V$

$\Rightarrow V_{CE} - I_C R_C - I_E R_E = 0$; $I_E \approx I_C$
 $9V - I_C (2k + 1k) = 0$
 $9V = I_C (3k)$
 $I_C = \frac{9V}{3k} = 3mA$

$V_{CE} = V_{CC} - I_C (R_C + R_E)$
 $V_{CE} = 9V - 3mA (3k) = 0V$

$5 (0.15F \cdot S) + 2F \cdot 1.8 =$

$R_{in} V_{BE} \cdot 2I = 20V \cdot 1.8 = 36V$

$1.11 - 1.11 = 0$

problem;

- a) I_C b) V_E c) V_{CE}
 d) V_{CE} e) V_B f) R_1

Hence, $R_2 = 6.8 \text{ k}\Omega$

$$R_E = 2.7 \text{ k}\Omega$$

$$R_E = 1.2 \text{ k}\Omega$$

$$V_E = 0.6 \text{ V} + I_E(1 + \beta) = 30 \text{ V} - 159 \text{ mV} - 15 \text{ V}$$

$$\beta = 100 \quad \frac{30 \text{ V} - 15 \text{ V}}{30 \text{ V} - 15 \text{ V}} = 100$$

$$I_B = 20 \mu\text{A} \quad \frac{30 \text{ V} - 15 \text{ V}}{30 \text{ V} - 15 \text{ V}} = 100$$

$$\text{a) } \beta = \frac{I_C}{I_B} \Rightarrow I_C = \beta I_B = (100) \times 20 \times 10^{-3} = 2 \text{ mA}$$

$$\text{b) } I_E = I_B + I_C = 2.02 \text{ mA} \quad \text{Ans}$$

$$V_E = 2.02 \times 1.2 = 2.424 \text{ V}$$

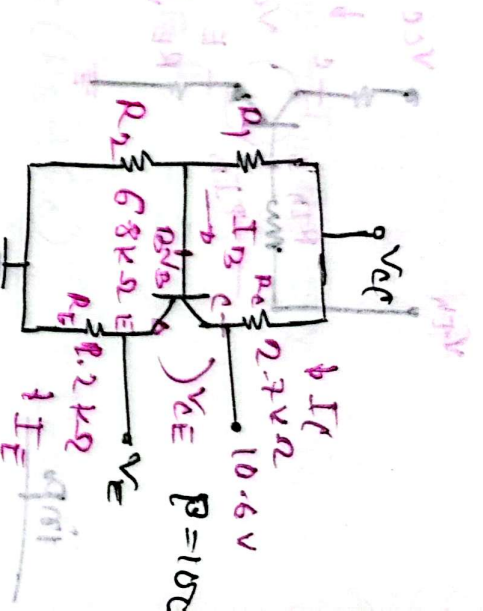
$$\text{d) } V_{CE} = V_{CC} - V_{BE} - I_E R_E = 10.6 \text{ V} - 0.6 \text{ V} - 2.424 \text{ V} = 8.176 \text{ V}$$

$$\text{e) } V_{CE} = V_{CE} + I_E(R_E + R_E) I_E = 30 \text{ V}$$

$$= 8.176 + (2.7 + 1.2) 2$$

$$\therefore V_{CE} = 15.976 \text{ V} \quad \text{Ans}$$

$$\text{Ans} \quad V_{CE} - I_E R_E = 10.6$$



Thermal Runaway in Transistors

I_{CBO} → reverse saturation current → depend on minority carrier

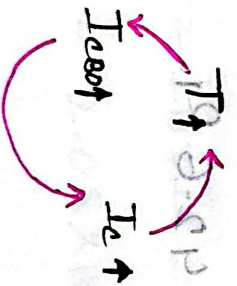
$T \uparrow \Rightarrow$ conc. of minority chrg. carrier $\uparrow \Rightarrow I_{CBO} \uparrow \Rightarrow$

$$S_o, \uparrow I_e = \beta I_B + (\beta + 1) I_{CBO} \quad , \quad = 20V - 3.5V \quad (t$$

$$I_{CBO} \propto T^{1.5} \quad \therefore \quad \frac{I_{CBO} - 25^{\circ}C}{I_{CBO} \propto T^{1.5}} = \frac{20V - 3.5V}{20V} = 11.9 \quad \leftarrow$$

$$\Rightarrow 10^{\circ}C \uparrow T \Rightarrow \text{double } I_{CBO} = \frac{24.019}{29.119} \text{ (won)}$$

$$\Rightarrow \text{Flow of } I_e \rightarrow \text{prod. heat} \rightarrow T \uparrow \Rightarrow I_{CBO} \uparrow \leftarrow$$



→ what is thermal runaway?

→ mitigating of I_{CBO} " $\Rightarrow$ negative feedback ($I_{CBO} \uparrow \Rightarrow I_B \downarrow \Rightarrow I_{CBO} \downarrow$)

→ (ii) Heat sinks. $215^{\circ}C \rightarrow 19^{\circ}C$

Bias stabilization and stability factor

stabilization: The process of making operating point independent of temperature changes and variation in transistor parameters is known as stabilization.

$$Q\text{-pt} \pm (V_{CE}, I_{CQ})$$

$$Q\text{-pt} \equiv (I_{CQ}, V_{CEQ})$$

→ causes of unstabilization:

Q-pt will shift $\rightarrow I_C \uparrow \downarrow$

i) change in β ;

$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

ii) change in T ; $T \uparrow \Rightarrow I_{CBO} \uparrow$ ($10^\circ \uparrow T \Rightarrow \text{double } I_{CBO}$)

→ V_{BE} decrease. 2-5 mV/°C $\uparrow T$

stability factor: The rate of change of collector I_C current with respect to the leakage current I_{CBO} of a constant input voltage V_{BE} and amplification factor β is called stability factor.

$$S = \left. \frac{dI_C}{dI_{CBO}} \right|_{\text{at const. } V_{BE} \text{ \& } \beta}$$

$$S' = \left. \frac{dI_C}{dV_{BE}} \right|_{\text{at const. } I_{CBO} \text{ \& } \beta}$$

$$S'' = \left. \frac{dI_C}{d\beta} \right|_{\text{at const. } V_{BE} \text{ \& } I_{CBO}}$$

$S \downarrow$ $\rightarrow$ more stable

if β is small (i)

$$I_{CBO} (1 + \beta) + \Delta I_C = \beta I_C$$

if β is small (ii)

$$V_{BE} = V_{BE} + \Delta V_{BE}$$

Stability factor for fixed Bias config

step 1: (KVL in loop) At (i) mps. h.b

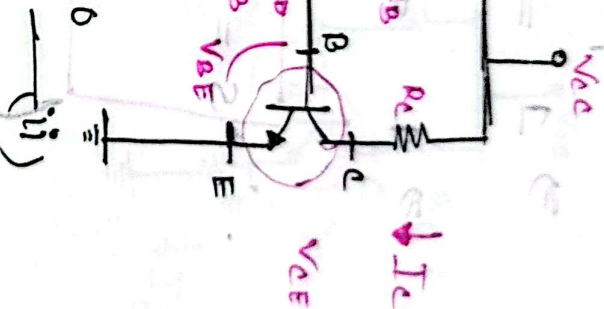
$$V_{CC} - I_B R_B - V_{BE} = 0 \quad (i) \text{ mps. h.b}$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Step 2:

$$I_E = \beta I_B + (\beta + 1) I_{CBO}$$

$$\Rightarrow I_E = \frac{\beta (V_{CC} - V_{BE})}{R_B} + (\beta + 1) I_{CBO}$$



step 3: diff. eq (ii) with I_E at const V_{BE} & β

$$S = \frac{dI_E}{dI_{CBO}}$$

$$; V_{BE} \& \beta, V_{CC}, R_B \rightarrow \text{const}$$

$$\frac{dI_E}{dI_{CBO}} = \frac{\beta}{R_B} \cdot \frac{d}{dI_{CBO}} (V_{CC} - V_{BE}) + (\beta + 1) \frac{d}{dI_{CBO}} (I_{CBO})$$

$$\Rightarrow 1 = 0 + (\beta + 1) \cdot \frac{1}{S}$$

$$\therefore \boxed{S = \beta + 1}$$

General eqn. set:

diff. eqn (i) with I_E eqn (iv) : eqn 2

$$\Rightarrow 1 = \beta \frac{dI_B}{dI_E} + (\beta + 1) \frac{dI_{CBO}}{dI_E} - 0.99 \cdot 0.1 = 0.1$$

$$\Rightarrow 1 = \beta \frac{dI_B}{dI_E} + \frac{(\beta + 1) \cdot 0.01}{0.1} = 0.1$$

$$S = \frac{(\beta + 1) - 0.99 \cdot 0.1 (1 + \beta) + 0.1 \beta}{0.1 (1 + \beta)} = 0.1$$

$$(q \beta E_{BV} + 0.99 \frac{\beta + 1}{1 - \beta} I_E) = (\beta + 1) I_E \cdot 0.1$$

Now,

$$\frac{dI_E}{dI_B} = \frac{\beta}{1 + \beta} \frac{dI_E}{dV_{BE}} \cdot \frac{dV_{BE}}{dI_B} \cdot \frac{dI_B}{dI_E} = \frac{\beta}{1 + \beta} \cdot \frac{dI_B}{dI_E} = 2$$

$$I_E = \beta \frac{(V_{EC} - V_{BE})}{R_B} + (\beta + 1) I_{CBO} + (i_i)$$

$$\Rightarrow 1 \neq \frac{\beta}{R_B} \left(0 - \frac{dV_{BE}}{dI_E} \right) + 0.1 = 2$$

$$\Rightarrow 1 = -\frac{\beta}{R_B} \left(-\frac{1}{S'} \right)$$

$$\Rightarrow \frac{1}{S'} = -\frac{\beta}{R_B} \Rightarrow S' = -\frac{R_B}{\beta}$$

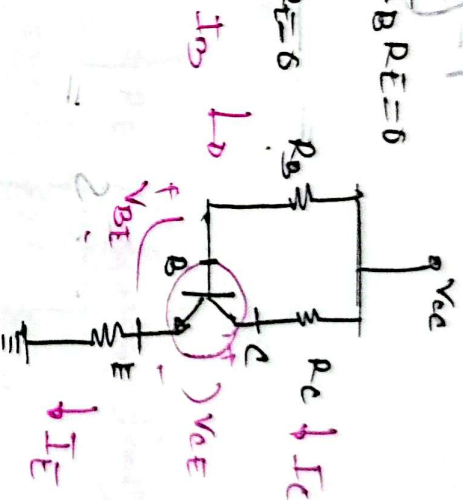
Sf. for emitter bias. Confi app. 03B

$$S1: V_{ce} - I_B R_B - V_{be} - I_E R_E = 0 \quad (1+g) = 2$$

$$\Rightarrow V_{ce} - I_B R_B - V_{be} - I_E R_E = 0$$

$$\Rightarrow V_{ce} - I_B (R_B + R_E) - V_{be} - I_E R_E = 0$$

$$\therefore I_B = \frac{V_{ce} - V_{be} - I_E R_E}{(R_B + R_E) (1+g)}$$



step 2 :

$$I_E = \beta I_B + (\beta + 1) I_{EBO}$$

$$\Rightarrow I_E = \beta \left(\frac{V_{ce} - V_{be} - I_E R_E}{R_B + R_E} \right) + (\beta + 1) I_{EBO}$$

step 3:

$$\frac{dI_E}{dI_E} = \frac{\beta}{R_B + R_E} (0 - 0 + R_E) + (\beta + 1) \frac{dI_{EBO}}{dI_E}$$

$$\Rightarrow 1 = \frac{I_{EBO} \cdot \beta}{R_B + R_E} + (\beta + 1) \cdot \frac{1}{S}$$

$$\Rightarrow \frac{(\beta + 1)}{S} = 1 + \frac{R_E \cdot \beta}{R_B + R_E} = \frac{R_B + R_E + \beta R_E}{R_B + R_E}$$

$$\Rightarrow S = (\beta + 1) \cdot \frac{R_B + R_E}{R_B + (\beta + 1) R_E}$$

$$\frac{(13+1)}{1 - \left(\frac{dT_0}{dT_c}\right)^2} \beta$$

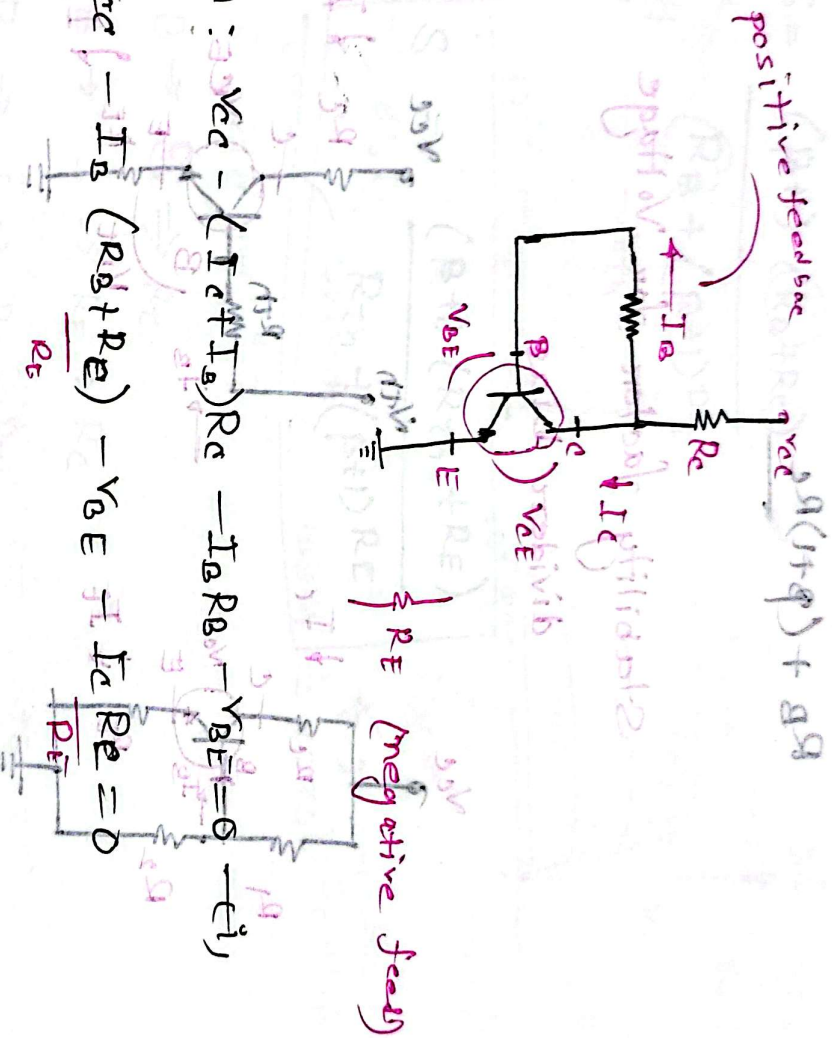


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SF for collector feedback biasing



step 1: $V_{ce} = (I_c + I_b)R_c - I_b R_B - V_{BE} = 0$

$\Rightarrow V_{ce} - I_b (R_B + R_c) - V_{BE} = 0$

$$I_B = \frac{V_{ce} - V_{BE} - I_c R_c}{R_B + R_c}$$

step 2: $I_c = \beta I_B + (\beta + 1)I_{CBO}$

$$\Rightarrow I_c = \beta \left(\frac{V_{ce} - V_{BE} - I_c R_c}{R_B + R_c} \right) + (\beta + 1)I_{CBO}$$

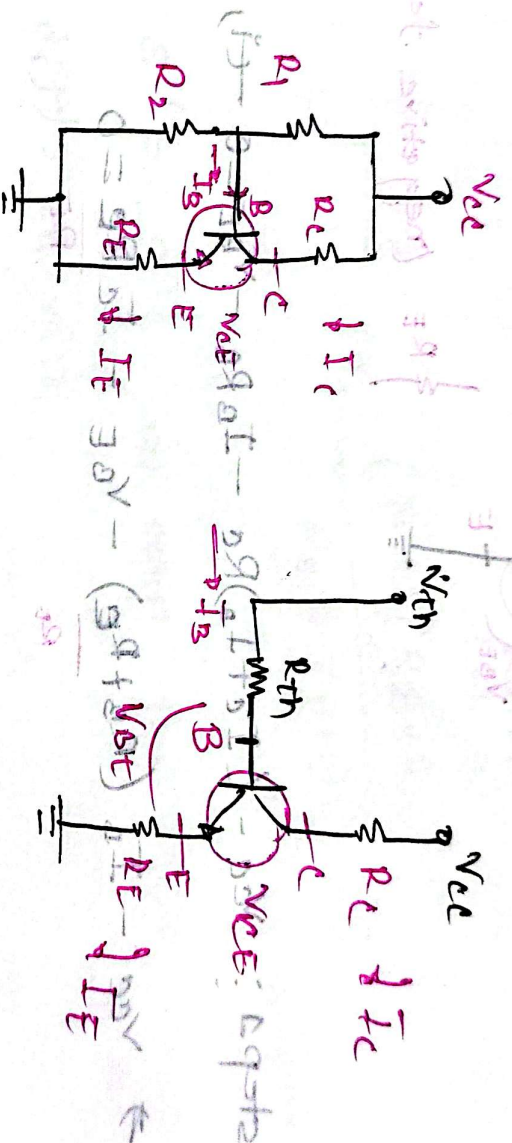
step 3: diff. eqn (ii) with res. R_c

$$\Rightarrow 1 \neq \frac{\beta}{\beta + 1} (-R_c) + \frac{(\beta + 1)}{S}$$

$$S = \frac{(\beta + 1)(R_B + R_c)}{R_B + (\beta + 1)R_c}$$

$$S' = \frac{\frac{1}{\beta}}{R_B + (\beta+1)R_E}$$

Stability factor for Voltage divider bias



$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}, \quad V_{Th} = \frac{R_2 V_{cc}}{R_1 + R_2}$$

$$S_1 \neq I_B = I_C \quad \text{and} \quad S_2 = I_C = I_E$$

$$(ii) \text{ --- } \text{Stability factor } S' = \frac{1}{\beta + 1} \left(1 + \frac{R_{Th}}{R_E} \right) \quad \text{and} \quad I_{CQ} = I_E$$

S3 : diff. I_C keeping β and V_{BE} const

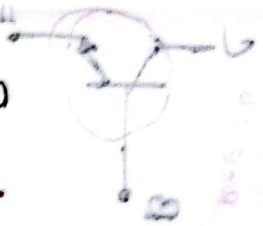
Stability factor (ii) $S' = \frac{1}{\beta + 1} \left(1 + \frac{R_{Th}}{R_E} \right)$

$$\frac{(1+\beta)}{2} + (1+\beta) \frac{R_{Th}}{R_E} \neq 1$$

$$S' = \frac{1}{\beta + 1} \left(1 + \frac{R_{Th}}{R_E} \right) = 1$$

for emitter bias

$$S = \frac{(\beta+1)(R_B + R_E)}{R_B + (\beta+1)R_E}$$



S_0 ,

$$S = \frac{(\beta+1)(R_{TH} + R_E)}{R_{TH} + (\beta+1)R_E}$$

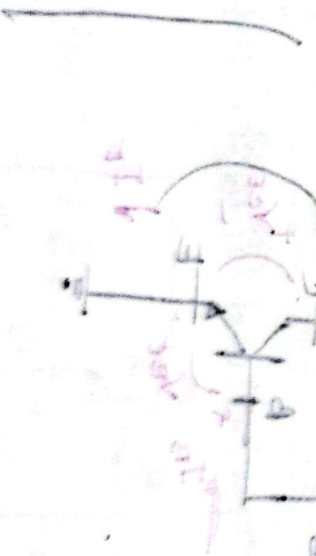
$$E_m B \rightarrow R \Rightarrow R_E$$

$$C.B \rightarrow S \Rightarrow R_E \rightarrow R_C$$

$$V.B \rightarrow S \Rightarrow R_B \rightarrow R_{TH}$$

$$S = \frac{\beta}{R_{TH} + (\beta+1)R_E}$$

$$0.39 \times 10^{-3} - 0.05V$$

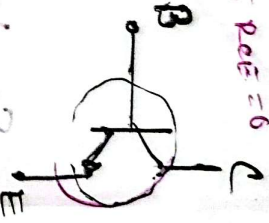
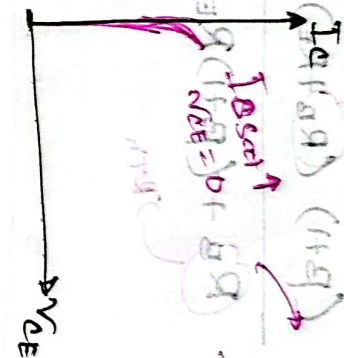
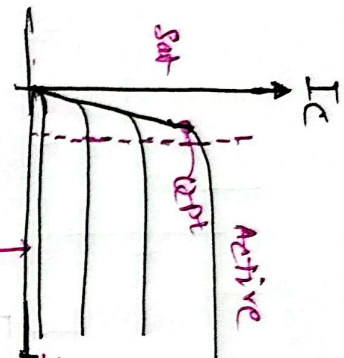


$$S = S(I_{CQ})$$

$$S' = S(V_{BE})$$

Transistor Saturation $\rightarrow V_{CE} = V_{CE}$

$$I_{C_{sat}} = I_{E_{sat}} = I_{B_{sat}} = I_{B_{sat}} = I_{B_{sat}}$$



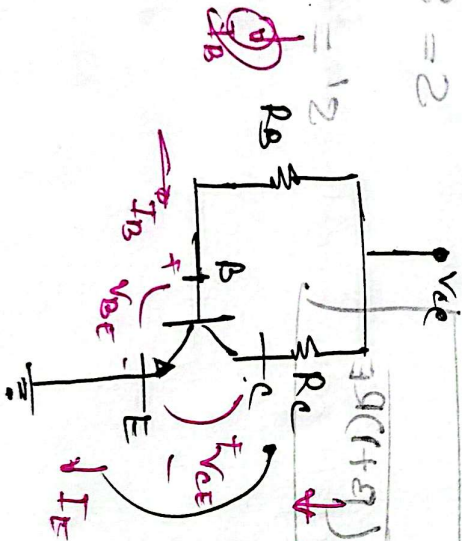
$$I_{C_{max}} \leq I_{C_{sat}}$$

Common Emi

saturation for fixed bias

$$(100\%)^2 = 2$$

$$(30\%)^2 = 12$$



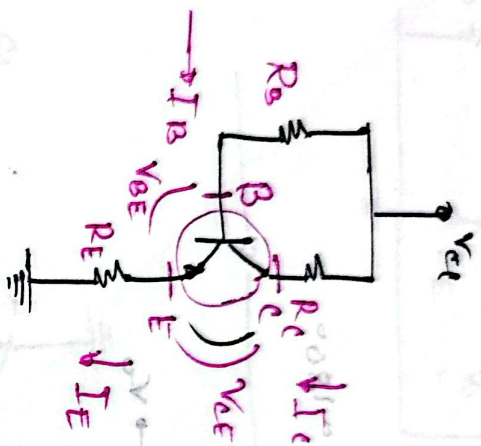
$$I_{C_{sat}} = I_{E_{sat}} = I_{B_{sat}} = I_{B_{sat}} = I_{B_{sat}}$$

$$I_{C_{sat}} = I_{E_{sat}} = I_{B_{sat}} = I_{B_{sat}} = I_{B_{sat}}$$

$$I_{C_{sat}} = I_{E_{sat}} = I_{B_{sat}} = I_{B_{sat}} = I_{B_{sat}}$$

$$I_{C_{sat}} = I_{E_{sat}} = I_{B_{sat}} = I_{B_{sat}} = I_{B_{sat}}$$

Emitter bias Conf.



$$I_{sat} = I_{cmax} = \frac{V_{ce}}{R_c + R_E}$$

$$V_{ce} - I_{sat} R_c - I_E R_E = 0$$

$$V_{ce} - I_{sat} R_c - I_E R_E = 0$$

$$\Rightarrow I_{sat} = \frac{V_{ce}}{R_c + R_E}$$

$$V_{ce} = 30V - 1V = 29V$$

$$I \approx \frac{\partial V}{\partial V} = 1V$$

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (1 + \beta) R_E}$$

$$R_B + (1 + \beta) R_E = \frac{V_{EE} - V_{BE}}{I_B}$$

